INCH - POUND
MIL-S-83505/1E
25 September 1991
SUPERSEDING
MIL-S-83505/1D
30 November 1987

MILITARY SPECIFICATION SHEET

SOCKETS (LEAD, ELECTRONIC COMPONENTS) (TYPE I, SOLDERLESS WRAP)

This specification is approved for use by all Departments and Agencies of the department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-83505

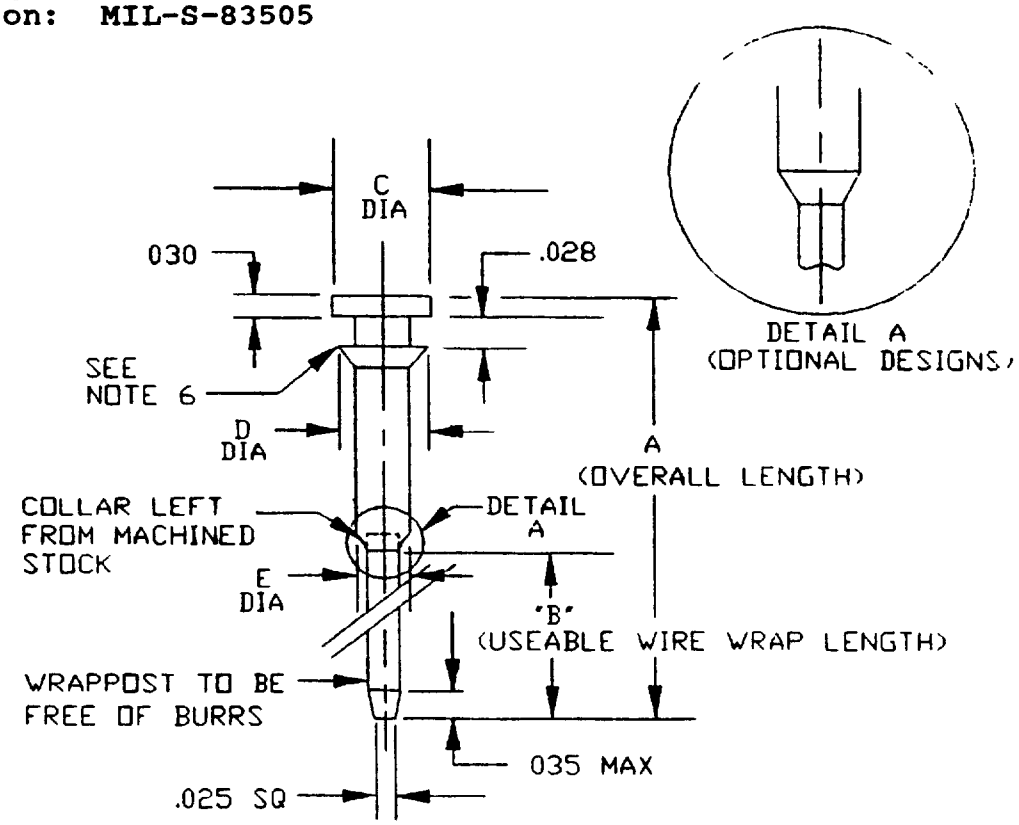


FIGURE 1. Socket, lead, (type I) solderless wrap.

Ⓔ denotes changes

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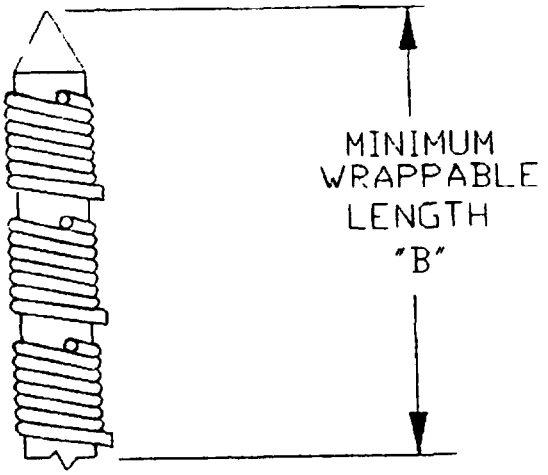
Dash number	A ±.010	B ±.010	C Dia	D Dia ±.002	E Dia
-001 -002	.703	.475	.072	.060	.052
-003 -004	.563	.335	.072	.060	.052
-005 -006	.453	.225	.072	.060	.052
-007 -008	.812	.475	.085	.082	.073
-009 -010	.703	.475	.063 ±.003	.060	.052
-011 -012	.563	.335	.063 ±.003	.060	.052
-013 -014	.453	.225	.063 ±.003	.060	.052

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.0012	0.030	.025	0.64	.073	1.85	.281	7.14
.0015	0.038	.028	0.71	.076	1.93	.335	8.51
.002	0.05	.030	0.76	.082	2.08	.370	9.40
.003	0.08	.0325	0.826	.085	2.16	.435	11.51
.005	0.13	.0355	0.902	.125	3.18	.475	12.07
.010	0.25	.046	1.17	.156	3.96	.510	12.95
.016	0.41	.052	1.32	.175	4.44	.563	14.30
.020	0.51	.055	1.40	.225	5.71	.703	17.86
.021	0.53	.060	1.52	.251	6.38	.812	20.62
.022	0.56	.072	1.83	.260	6.60		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified the tolerance for three place decimals is ±.005.
Ⓔ 4. Part is to be free of burrs or sharp edges except as noted.
5. Dimensions apply after plating.
Ⓔ 6. For retention design requirements see MIL-S-83505, figure 2.

FIGURE 1. Socket, lead, (type I) solderless wrap - Continued.

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Ⓔ Minimum wrappable length, -B, inches (see note 5)

Number of wrapped connections	Wire gauge (AWG)						
	30	28	26	24	22	20	18
1	185 (4.70)	219 (5.56)	226 (5 74)	258 (6.55)	303 (7 70)	327 (8 31)	.391 (9 93)
2	.320 (8 13)	388 (9.86)	402 (10.21	466 (11 84)			
3	455 (11.56)						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified the tolerance is ±.005 (0.13 mm).
5. Minimum wrappable length shall have the wrappost geometry in accordance with table IA of MIL-STD-1130.

FIGURE 2. Wrapppost configurations and wire gauge length.

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REQUIREMENTS:

Qualification:

Group qualification: Shall be as specified in table I.

Dimensions and configuration: Shall be in accordance with MIL-STD-1130, figures 1 and 2 and table II.

Ⓔ TABLE I. Group qualification.

Dash number	Qualification tests
-001, -002, -009, and -010 (as applicable)	Require all qualification tests of MIL-S-83505.
-003 through -006 and -011 through -014 (as applicable)	Visual an mechanical tests of MIL-S-83505 and qualification of -001, or -002, or -009, or -010.
-007 and -008 (as applicable)	Groups 1, 2, and 5 of qualification table of MIL-S-83505 and qualification of -001, or -002, or -009, or -010.

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Ⓔ TABLE II. Interface dimensions and finish.

Dash number	Recommended mounting hole dia ±.001	Accepts pin		Sleeve plating
		Length	Diameter	
-001 and -009	.055	.175	.021	Gold
-002 and -010		.125	.016	Tin/lead
-003 and -011	.055	.175	.021	Gold
-004 and -012		.125	.016	Tin/lead
-005 and -013	.055	.175	.021	Gold
-006 and -014		.125	.016	Tin/lead
-007	.076	.281	.030	Gold
-008		.156	.020	Tin/lead

Engagement (mating) and separation (unmating) forces shall be in accordance with MIL-S-83505 except as specified in table III.

Ⓔ TABLE III. Engagement (mating) force.

Dash numbers	Test gauge B	Mating force (max) (lbs)
-001 and -009	.125	1.25
-002 and -010		
-003 and -011	.125	1.25
-004 and -012		
-005 and -013	.125	1.25
-006 and -014		
-007	.156	4.00
-008		

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Terminals: Type I (solderless wrap).

Mounting: Use .046 inch (1.16 mm) minimum thickness mounting board.

Test mounting:

Use glass epoxy board, type G10 material for mounting sockets.

Ⓔ Part or Identifying Number (PIN): M83505/1- (dash number from table II).

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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